Bulletin PD-2.397 rev. G 07/04

International
IOR Rectifier

10BQ040

SCHOTTKY RECTIFIER 1 Amp

$I_{F(AV)}$ = 1 Amp
V_R = 40V

Major Ratings and Characteristics

Characteristics	10BQ040	Units
$I_{F(AV)}$ Rectangular waveform	1.0	A
V_{RRM}	40	V
I_{FSM} @tp = 5 µs sine	430	A
V_F @1.0 Apk, T_J=125°C	0.49	V
T_J range	- 55 to 150	°C

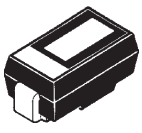
Description/ Features

The 10BQ040 surface-mount Schottky rectifier has been designed for applications requiring low forward drop and very small foot prints on PC boards. Typical applications are in disk drives, switching power supplies, converters, free-wheeling diodes, battery charging, and reverse battery protection.

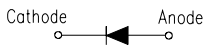
- Small foot print, surface mountable
- Low forward voltage drop
- High frequency operation
- Guard ring for enhanced ruggedness and long term reliability

Case Styles

10BQ040



SMB



Voltage Ratings

	Part number	10BQ040
V_R	Max. DC Reverse Voltage (V)	40
V_{RWM}	Max. Working Peak Reverse Voltage (V)	

Absolute Maximum Ratings

	Parameters	10BQ	Units	Conditions	
$I_{F(AV)}$	Max. Average Forward Current	1.0	A	50% duty cycle @ T_L = 112 °C, rectangular wave form	
I_{FSM}	Max. Peak One Cycle Non-Repetitive	430	A	5µs Sine or 3µs Rect. pulse	Following any rated load condition and with rated V_{RRM} applied
	Surge Current	45		10ms Sine or 6ms Rect. pulse	
E_{AS}	Non- Repetitive Avalanche Energy	3.0	mJ	T_J = 25 °C, I_{AS} = 1A, L = 6mH	
I_{AR}	Repetitive Avalanche Current	1.0	A	Current decaying linearly to zero in 1 µsec Frequency limited by T_J max. Va = 1.5 x Vr typical	

Electrical Specifications

	Parameters	10BQ	Units	Conditions	
V_{FM}	Max. Forward Voltage Drop (1)	0.53	V	@ 1A	T_J = 25 °C
	* See Fig. 1	0.70	V	@ 2A	
		0.49	V	@ 1A	T_J = 125 °C
		0.64	V	@ 2A	
I_{RM}	Max. Reverse Leakage Current (1)	0.1	mA	T_J = 25 °C	V_R = rated V_R
	* See Fig. 2	4	mA	T_J = 125 °C	
C_T	Typical Junction Capacitance	80	pF	V_R = 5V_{DC}, (test signal range 100kHz to 1MHz) 25°C	
L_S	Typical Series Inductance	2.0	nH	Measured lead to lead 5mm from package body	
dv/dt	Max. Volatge Rate of Charge (Rated V_R)	10000	V/ µs		

(1) Pulse Width < 300µs, Duty Cycle < 2%

Thermal-Mechanical Specifications

	Parameters	10BQ	Units	Conditions
T_J	Max. Junction Temperature Range (*)	- 55 to 150	°C	
T_{stg}	Max. Storage Temperature Range	- 55 to 150	°C	
R_{thJL}	Max. Thermal Resistance Junction to Lead (**)	36	°C/W	DC operation
R_{thJA}	Max. Thermal Resistance Junction to Ambient	80	°C/W	
wt	Approximate Weight	0.10 (0.003)	g (oz.)	
	Case Style	SMB		Similar DO-214AA
	Device Marking	IR1F		

(*) $\frac{dPtot}{dTj} < \frac{1}{Rth(j\text{-}a)}$ thermal runaway condition for a diode on its own heatsink

(**) Mounted 1 inch square PCB

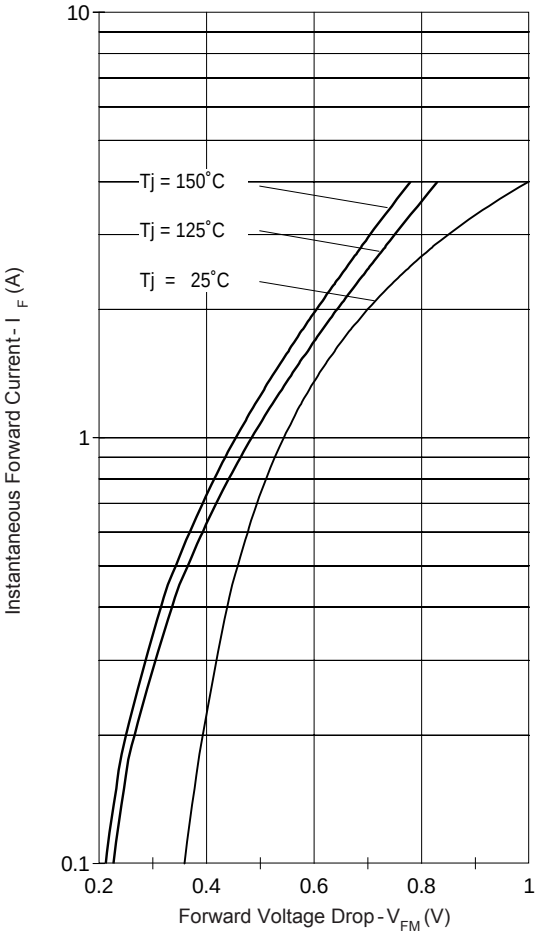


Fig. 1 - Maximum Forward Voltage Drop Characteristics

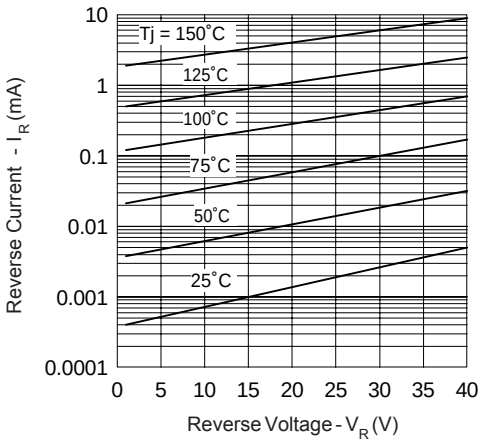


Fig. 2 - Typical Peak Reverse Current
Vs. Reverse Voltage

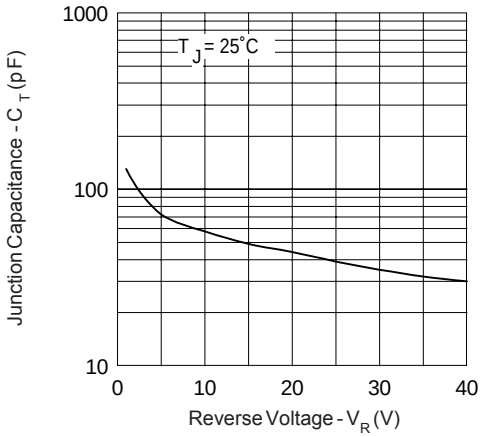


Fig. 3 - Typical Junction Capacitance
Vs. Reverse Voltage

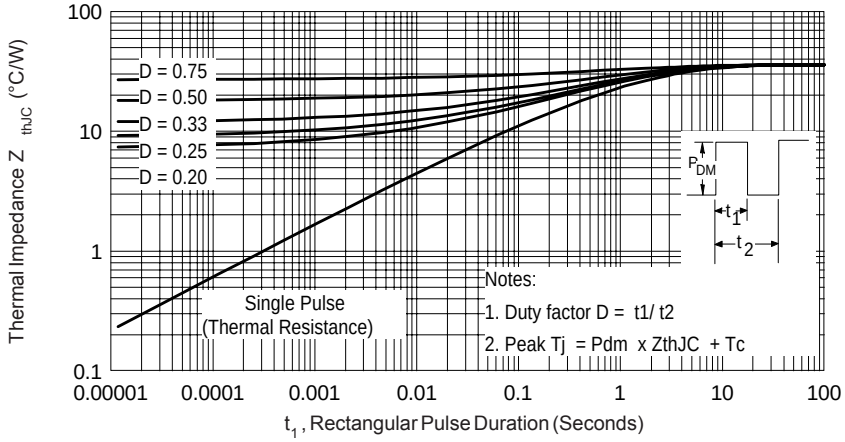


Fig. 4 - Max. Thermal Impedance Z_{thJC} Characteristics (Per Leg)

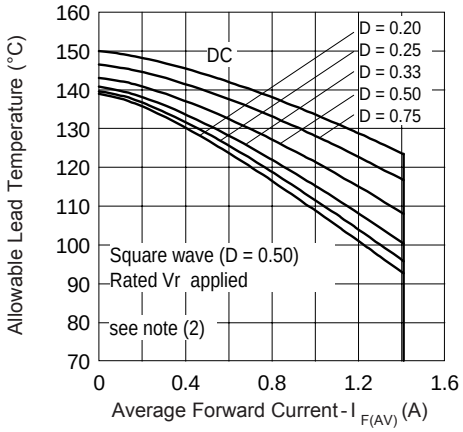


Fig. 4 - Maximum Average Forward Current
Vs. Allowable Lead Temperature

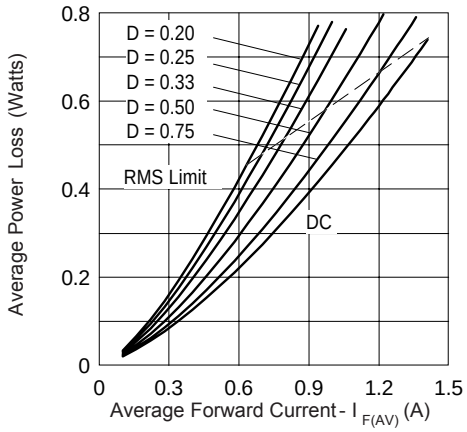


Fig. 5 - Maximum Average Forward Dissipation
Vs. Average Forward Current

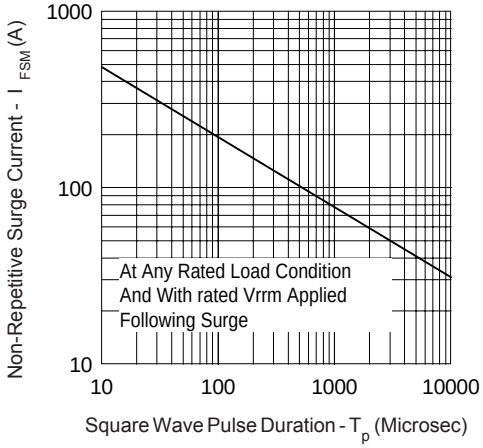
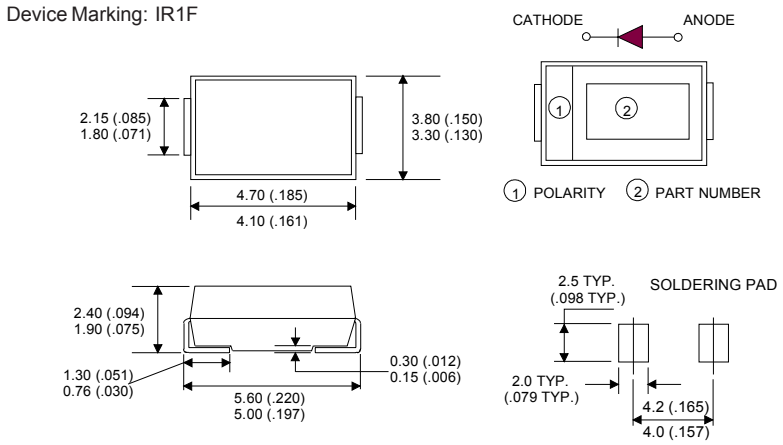


Fig. 6 - Maximum Peak Surge Forward Current Vs. Pulse Duration

(2) Formula used: $T_C = T_J - (Pd + Pd_{REV}) \times R_{thJC}$;
Pd = Forward Power Loss = $I_{F(AV)} \times V_{FM} @ (I_{F(AV)} / D)$ (see Fig. 6);
Pd_{REV} = Inverse Power Loss = $V_{R1} \times I_R (1 - D)$; $I_R @ V_{R1}$ = 80% rated V_R

Outline Table



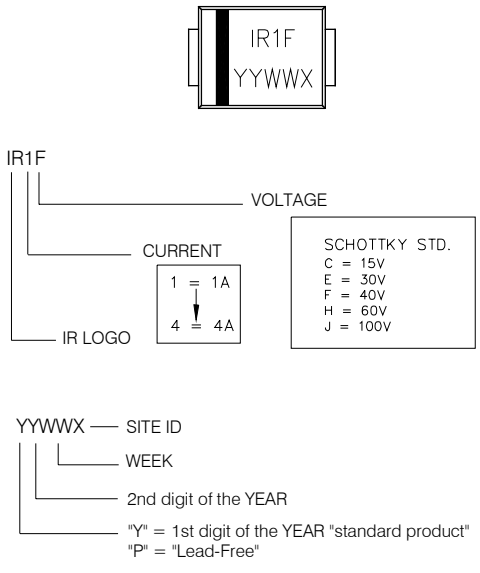
Outline SMB

Dimensions in millimeters and (inches)

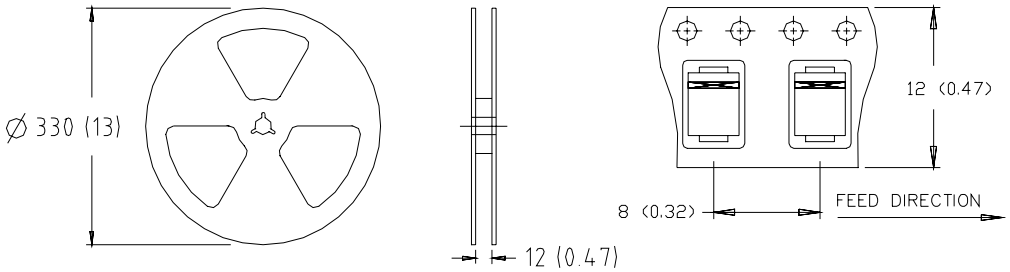
For recommended footprint and soldering techniques refer to application note #AN-994

Marking & Identification

Each device has 2 rows for identification. The first row designates the device as manufactured by International Rectifier, indicated by the letters "IR", and the Part Number (indicates the current, the voltage rating and Schottky Generation). The second row indicates the year, the week of manufacturing and the Site ID.



Tape & Reel Information



Dimensions in millimetres and (inches)

Ordering Information Table

Device Code

10	B	Q	040	TR	-
1	2	3	4	5	6

1	-	Current Rating
2	-	B = Single Lead Diode
3	-	Q = Schottky Q Series
4	-	Voltage Rating (040 = 40V)
5	-	• none = Box (1000 pieces)
		• TR = Tape & Reel (3000 pieces)
6		• none = Standard Production
		• PbF = Lead-Free

Data and specifications subject to change without notice.
This product has been designed and qualified for Industrial Level.
Qualification Standards can be found on IR's Web site.

International
IOR Rectifier

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